

Extension III Mathematics Activity Sheet 1

Recurrence relations and generating functions

Ideally, you should watch the associated YouTube video “Recurrence Relations and Generating Functions” to get a good introduction to the material in this Activity Sheet. A quick summary is below.

This Activity Sheet gets you to look at recurrence relations for sequences $a_0, a_1, \dots$. An example of such a recurrence relation is the one generating the Fibonacci numbers: $a_{n+1} = a_n + a_{n-1}$ with initial conditions $a_0 = a_1 = 1$. One basic tool for studying them will be the notion of the corresponding generating function $A(x) = \sum_{n=0}^{\infty} a_n x^n$. It is a useful bookkeeping device, in part, because the shifted sequence $0, a_0, a_1, \dots$ which can be thought of as $\{a_{n-1}\}$ has generating function $\sum a_{n-1} x^n = xA(x)$. For example, the recurrence relation for the Fibonacci numbers would imply that

$$A(x) - xA(x) - x^2A(x) = 1$$

as is easily seen on comparing coefficients of both sides. Thus we may use partial fractions to see

$$A(x) = \frac{1}{1-x-x^2} = \frac{1}{\sqrt{5}} \frac{1}{1-\alpha_+x} - \frac{1}{\sqrt{5}} \frac{1}{1-\alpha_-x}$$

where $\alpha_{\pm} = \frac{1 \pm \sqrt{5}}{2}$. We now use the sum of an infinite geometric progression formula to extract the coefficients of each fraction to give

$$a_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n+1}$$

We explore this technique in this Activity Sheet.

Afterwards, you might wish to look deeper as to why the method works by doing the Activity Sheets on Series and Power Series, and/or the Activity Sheet on Formal Power Series.

1. Consider the recurrence relation. given by

$$a_n = 2a_{n-1} - a_{n-2}, \quad \text{for } n \geq 2.$$

- (a) Suppose the first two terms of the sequence are $a_0 = 3$, $a_1 = 5$. Compute the next 3 terms.

- (b) Can you see a pattern and guess what the formula for a_n as a function of n is?
 - (c) Prove your guess in the previous part by induction.
 - (d) Suppose now that a_0, a_1 are some arbitrary fixed numbers. What types of sequences can you get satisfying the above recurrence relation? Find a satisfying reason why this is true. Hint: you have studied these types of sequences before.
 - (e) In Q8, you will want to revisit this question by using generating functions.
2. Suppose now the sequence $\{a_n\}_{n \in \mathbb{N}}$ satisfies the recurrence relation

$$a_n - 4a_{n-1} + 3a_{n-2} = 0.$$

with initial values $a_0 = 3, a_1 = 5$.

- (a) Find the next 3 terms of the sequence.
 - (b) Determine the associated generating function $A(x)$.
 - (c) Using partial fractions on $A(x)$, give a closed formula for a_n .
3. Let b, c, d be real numbers. Consider the sequence
- $$a_n = b + c2^n + d5^n, \text{ for } n \in \mathbb{N}$$
- (a) If $b = c = d = 1$, find a closed form for the generating function $A(x)$.
 - (b) For arbitrary b, c, d , find a recurrence relation that $\{a_n\}_{n \in \mathbb{N}}$ satisfies.
4. Suppose that the sequence $\{a_n\}_{n \in \mathbb{N}}$ is 3-periodic in the sense that $a_n = a_{n-3}$ for $n \geq 3$. Show that there are complex numbers b, c, d such that

$$a_n = b + ce^{2\pi in/3} + de^{-2\pi in/3}.$$

5. At the beginning of every year, Grothendieck deposits \$1 into Bank SuperRisky. At the end of the year, the account earns 50% interest. If a_n denotes the amount in the account at the end of the n -th year, determine a_n . (Assume here for definiteness that $a_0 = 0$.)

6. Recurrence relations are an important tool to solve “counting” problems. Here “counting” is mathematical jargon for finding the number of elements in some set. We explore here an example counting “reduced words”.

We start with 3 symbols x, x^{-1}, y . A *word of length n* in these symbols is a string or expression $z_1 z_2 \dots z_n$ where the $z_1, \dots, z_n$ are some choice of the 3 symbols. For example $xyyx^{-1}xyx$ is a word of length 7 and by convention, there is a unique empty word of length 0. The symbology is chosen to suggest one can cancel adjacent x and x^{-1} to reduce the word to a shorter one. A word is *reduced* if there are no such adjacent x and x^{-1} in the word. Note $x^{-1}x$ is not reduced but xyx^{-1} is. Let a_n be the number of reduced words in x, x^{-1}, y of length n .

- (a) Find a_0, a_1 .
- (b) Show that the number of reduced words of length n for $n \geq 1$, where the last two symbols are the same, is a_{n-1} .
- (c) Show that the number of reduced words length $n, n \geq 2$ which end in yx or yx^{-1} is $2a_{n-2}$.
- (d) Show that the number of reduced words of length $n, n \geq 2$ which end in xy or $x^{-1}y$ is $a_{n-1} - a_{n-2}$.
- (e) Hence derived the recurrence relation

$$a_n = 2a_{n-1} + a_{n-2}, \quad n \geq 2.$$

- (f) Compute the generating function $A(x)$ and hence count the number of reduced words of length n i.e. find a_n .

7. This question will make more sense after you’ve seen the theory of (absolutely) convergent sequences and series. See how far you can get for now. The basic idea is that one would like to do “term by term differentiation” to get the beautiful formula (valid for $|x| < 1$)

$$\frac{1}{(1-x)^2} = \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{d}{dx} \sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} nx^{n-1} = \sum_{n=0}^{\infty} (n+1)x^n \quad (1)$$

It turns out that “term by term differentiation” is valid essentially whenever it makes sense. We will establish formula (1) by indirect means instead.

- (a) Let $B(x) = \sum_{n=0}^{\infty} (n+1)x^n$. The first step is to show that $B(x)$ converges for $|x| < 1$ and the key to this is that it does not shoot off to ∞ . Fix $0 < x < 1$. Pick first any y with $x < y < 1$ and positive integer N , large enough such that $\frac{n+1}{n}x < y$ for all $n \geq N$. Hence find a geometric progression g_n with common ratio y such that $(n+1)x^n < g_n$. Since the sum of the geometric progression with common ratio y converges, so does the sum $B(x)$.
 - (b) Show that $(1-x)B(x) = \sum_{n=0}^{\infty} x^n$ by expanding the LHS and manipulating the terms.
 - (c) Hence establish (1).
8. Use Q7 to redo Q1.
9. Make up your own recurrence relation questions that you can solve using generating functions. Can you make up some that you can't?

Partial answers, comments and hints

1. $a_n = 2n + 3$ in case (a). In general, you get arbitrary arithmetic progressions since you can re-write the recurrence relation as $a_n - a_{n-1} = a_{n-1} - a_{n-2}$.
2. $A(x) = \frac{3-7x}{1-4x+3x^2} = \frac{2}{1-x} + \frac{1}{1-3x}$ so $a_n = 2 + 3^n$
3. $A(x) = \frac{b}{1-x} + \frac{c}{1-2x} + \frac{d}{1-5x} = \frac{p(x)}{1-8x+17x^2-10x^3}$ for some polynomial $p(x)$ which we don't need to know. The denominator gives the recurrence relation

$$a_n - 8a_{n-1} + 17a_{n-2} - 10a_{n-3} = 0, \quad \text{for } n \geq 3.$$

4. The generating function in this case satisfies $(1-x^3)A(x) = p_0 + p_1x + p_2x^2$ for some constants p_0, p_1, p_2 . Now use the partial fraction expansion of $A(x)$ and the fact that the roots of $1-x^3$ are $x = 1, e^{2\pi i/3}, e^{-2\pi i/3}$.
5. The recurrence relation is now $a_{n+1} = \frac{3}{2}(a_n + 1)$ from which we see the generating function satisfies

$$(1 - \frac{3}{2}x)A(x) = \frac{3/2}{1-x}$$

so

$$A(x) = \frac{3}{(1-x)(2-3x)} = \frac{9}{2-3x} - \frac{3}{1-x}$$

from which we can compute a_n using GPs.

6. Note $a_0 = 1, a_1 = 3$. For part d), think about the reduced words ending in y and those ending in yy . $A(x) = \frac{1+x}{1-2x-x^2}$. Use partial fractions to find a_n .
7. Typical in this game is the fact that, coming up with the correct infinite sum formula for $\frac{1}{(1-x)^2}$ is not hard, the tricky part is the technical difficulty of ensuring the sum converges. If you think that one ought not need to involve such arguments, then you can set up formal power series instead as will be done in a later Activity Sheet.
8. Show that part (a) gives the generating function $A(x) = \frac{1}{1-x} + \frac{2}{(1-x)^2}$

Extension III Mathematics Series

This Activity Sheet looks at the summation of series, a technique necessary for manipulating generating functions and understanding how they work. It should serve as a quick primer in series. Do look at the associated videos on the webpage (see above for the link). You can look at standard calculus texts for a full treatment. If you have access to the UNSW MATH1231 Calculus notes, I've given references below. You should have seen the concepts below to complete this Activity Sheet. There is a lot more to learn about series, but one only needs a little bit to work with generating functions in enumerative combinatorics.

- Definition of a limit. Given a sequence $\{a_n\}_{n \in \mathbb{N}}$, we write $\lim_{n \rightarrow \infty} a_n = l$ if for every $\varepsilon > 0$, we can find a real number M such that

$$|a_n - l| < \varepsilon, \quad \text{whenever } n \geq M.$$

(see MATH1231 Calculus Notes: §4.3).

- Cauchy completeness. A sequence $\{a_n\}_{n \in \mathbb{N}}$ of complex numbers is *Cauchy* if for every $\varepsilon > 0$, we can find a real number M such that

$$|a_n - a_m| < \varepsilon, \quad \text{whenever } n \geq m \geq M.$$

Such sequences always converge.

- The series $\sum_{n=0}^{\infty} a_n$ *converges* to Σ if

$$\lim_{N \rightarrow \infty} \sum_{n=0}^N a_n = \Sigma.$$

If $\sum_{n=0}^{\infty} |a_n|$ converges then so does $\sum_{n=0}^{\infty} a_n$ by Cauchy completeness and we say this sum converges *absolutely*. (see MATH1231 Calculus Notes: §4.5.8).

- Suppose the series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ converge absolutely. Then the *Cauchy product series* $\sum_{n=0}^{\infty} c_n$ defined by $c_n = \sum_{j=0}^n a_j b_{n-j}$ also does and

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right)$$

(see MATH1231 Calculus Notes: §4.8).

- If $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r$, the *radius of convergence*, then the power series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for all $|x| < r$. Here we allow $r = \infty$ as a possibility. In particular we have the following *ratio test* for convergence, if $r > 1$, then $\sum_n a_n$ converges absolutely. (see MATH1231 Calculus Notes: §4.5.6).

Activities

1. Prove from definitions that $\lim_{n \rightarrow \infty} \frac{2n-1}{n+1} = 2$.
2. Suppose that the sums $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ converge to a and b respectively. If c is another number, then prove from the definition of limits that the sum $\sum_{n=0}^{\infty} ca_n$ converges to ca and $\sum_{n=0}^{\infty} (a_n + b_n)$ converges to $a + b$.
3. Show from the definition of the limit that the sum $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges as follows.
 - (a) Use a partial fraction expansion of the summands $\frac{1}{n(n+1)}$.
 - (b) Show that the partial sum $\sum_{n=1}^N \frac{1}{n(n+1)} = 1 - \frac{1}{N+1}$.
 - (c) Find the sum of the series. Use the definition of limits to prove from first principles that your answer gives the sum.
4. Show that the series $1 - 1 + 1 - 1 + 1 - 1 + \dots$ diverges, i.e. does not converge.
5. In this question, we look at an example of what's called the "integral test" for convergence. You can look this up in all the standard calculus texts or better still, try to infer from this example, how it works in general. We will show that the "harmonic series" $1 + \frac{1}{2} + \frac{1}{3} + \dots$ diverges by considering the function $f(x) = \frac{1}{x}$.
 - (a) Let R_n be the rectangle in the xy -plane defined by $n \leq x \leq n+1$, $0 \leq y \leq f(n)$. Show that the union of all these rectangles covers the area under the graph $y = f(x)$ defined by $1 \leq x, 0 \leq y \leq f(x)$.
 - (b) Conclude that

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \geq \int_1^n \frac{1}{x} dx.$$

- (c) By evaluating the integral in part b), show that the harmonic series diverges to infinity.
6. Use the technique of the last question to show that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.
 7. Show that Cauchy completeness of the reals implies the following result: given any increasing sequence $\{a_n\}_{n \in \mathbb{N}}$ which is bounded above say by b so $a_n \leq b$ for all n , then the sequence has a limit.
 8. Show that $\sum_{n=0}^{\infty} n!x^n$ converges absolutely only for $x = 0$.
 9. Let x be a real number with $|x| < 1$. Find the Cauchy product of the geometric series $1 + x + x^2 \dots$ with itself. Hence sum the resulting series.
 10. Use the ratio test to show that $\sum_{n=0}^{\infty} \frac{n^3}{2^n - 1}$ converges absolutely.
 11. Find the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{x^n}{n^2 2^n}$.
 12. Look up the *root test* for convergence in any standard calculus text. See if you can prove it.

Partial answers, comments and hints

1. Note $a_n := \frac{2n-1}{n+1} = 2 - \frac{3}{n+1}$. Thus for any $\varepsilon > 0$, we will have $|a_n - 2| < \varepsilon$ whenever $n > M$ where $M = \frac{3}{\varepsilon} - 1$. We can also pick bigger values for M .
2. We may assume $c \neq 0$ (make sure you know why!). By definition of the limit, for any $\varepsilon > 0$, there is some M such that

$$\left| a - \sum_{n=0}^N a_n \right| < \frac{\varepsilon}{|c|} \quad \text{whenever } N > M.$$

Now multiply this inequality by $|c|$. The proof for the sum of the series is similar.

3. This is the classic example of a “telescoping sum”. It is

$$\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots$$

Note that there are lots of cancellation of terms

4. The partial sums alternate between 1 and 0. This sequence does not tend to a limit.
5. The rectangles R_n in this question are sometimes called rectangles circumscribed about the graph of $f(x) = \frac{1}{x}$ so the integral gives a lower bound on the sum. The sum is an example of a Riemann sum used in the definition of the integral.
6. In this case, consider the graph of $f(x) = \frac{1}{x^2}$. The integral of $f(x)$ is meant to give an upper bound on the partial sums (and hence sum). One thus uses inscribed rectangles instead i.e. those of form $n \leq x \leq n+1$, $0 \leq y \leq \frac{1}{(n+1)^2}$.
7. Otherwise, there exists an $\varepsilon > 0$ such that for any a_m , we can find $n > m$ such that $a_n - a_m > \varepsilon$. Similarly $l > n$ such that $a_l - a_n > \varepsilon$ and so forth so the sequence increases without bound.
8. Note for $x \neq 0$, we have $\lim_{n \rightarrow \infty} n!|x|^n = \infty$.
9. $\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n$.
10. Just note $\lim_{n \rightarrow \infty} \frac{n^3 2^{-n}}{(n+1)^3 2^{-(n+1)}} = 2 > 1$.
11. $r = \lim_{n \rightarrow \infty} \frac{n^{-2} 2^{-n}}{(n+1)^{-2} 2^{-(n+1)}} = 2$.
12. The proof is included in most standard calculus texts, but it is similar to that of the ratio test so it's a good exercise to prove it.

Extension III Mathematics Activity Sheet Power Series

This activity sheet serves as a quick primer on power series and Maclaurin series. It should be enough to allow you to study questions involving generating functions in basic counting problems. Ideally, you should have done the Activity Sheet on Series before completing this one. Do look at the accompanying video on the webpage above to make sure you know the facts and notions below. Alternatively, you can find the material in most standard calculus texts (there are references to UNSW's Calculus notes below). We assume the following notation. Suppose $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for $|x| < r_a$ to $A(x)$ and $\sum_{n=0}^{\infty} b_n x^n$ converges absolutely for $|x| < r_b$ to $B(x)$. We assume $r_a < r_b$.

- The *Cauchy product series*

$$\sum_{n=0}^{\infty} \left(\sum_{i=0}^n a_i b_{n-i} \right) x^n = A(x)B(x)$$

converges absolutely for $|x| < r_a$ to the product $A(x)B(x)$. (See MATH1231 Calculus Notes §4.8).

- Term-by-term differentiation. $A(x)$ is differentiable for $|x| < r_a$ and we have the following absolutely convergent power series and formula

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = A'(x).$$

(See MATH1231 Calculus Notes §4.8).

- Maclaurin series. Let $f(x)$ be an infinitely differentiable real-valued function. Its *Maclaurin series* is $\sum_{n=0}^{\infty} f_n x^n$ where $f_n = \frac{f^{(n)}(0)}{n!}$. Here $f^{(n)}$ means the n -th derivative of f . (See MATH1231 Calculus Notes §4.6).
- Taylor's theorem. To see if the Maclaurin series represents $f(x)$ we can use the integral form for the remainder $R_n(x) = f(x) - \sum_{j=0}^n f_j x^j$. It applies if f has a continuous $(n+1)$ -st derivative at 0.

$$R_n(x) = \frac{1}{n!} \int_0^x f^{(n+1)}(t)(x-t)^n dt.$$

(See MATH1231 Calculus Notes §4.2).

- In particular we have for all $x \in \mathbb{R}$ (actually even in $\mathbb{C}$)

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

After this Activity Sheet, you might want to look at the Activity Sheet on formal power series which gives a purely algebraic approach to this material, or see how to use power series in counting problems.

Activities

1. Suppose for simplicity that $A(x) = \sum_{n=0}^{\infty} a_n x^n$ where $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r > 0$. Show that the Maclaurin series for $A(x)$ is itself. In particular, this means for example, that since we know $\frac{1}{1-x} = 1 + x + \dots$ for $|x| < 1$, then the geometric series on the RHS is the Maclaurin series for $\frac{1}{1-x}$.
2. Here we examine the Maclaurin series for $y = e^{2x}$.
 - (a) Compute the Maclaurin series first directly from the definition, and then by taking the Cauchy product of the Maclaurin series for e^x with itself.
 - (b) Show they coincide with the answer given by substituting $2x$ for x in the Maclaurin series for e^x .
 - (c) Show $y = e^{2x}$ satisfies the differential equation $\frac{dy}{dx} = 2y$ by using the Maclaurin series and term-by-term differentiation.
3.
 - (a) Determine the Maclaurin series for $\cos x$ and $\sin x$ and show using Taylor's theorem that they represent the trigonometric functions for all real x .
 - (b) Verify from the Maclaurin series that $\cos x$ and $\sin x$ are even and odd respectively.
 - (c) Show by substitution into the Maclaurin series (which are actually valid for complex numbers as well) that $e^{iy} = \cos y + i \sin y$ for all real y .

4. Just as you can differentiate convergent power series term-by-term, you can do the same with integration. Suppose for simplicity that $A(x) = \sum_{n=0}^{\infty} a_n x^n$ where $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r > 0$.
- (a) Show that $B(x) = \sum_{n=0}^{\infty} \frac{1}{n+1} a_n x^{n+1}$ also converges absolutely for $|x| < r$.
- (b) Apply term-by-term differentiation on $B(x)$ to show that

$$B(x) = \int_0^x A(t) dt.$$

5. Use term-by-term integration of $\frac{1}{1-x}$ to find the Maclaurin series for $\ln(1-x)$.
6. There is a generalisation of the binomial formula called the binomial series. For $\alpha \in \mathbb{R}$ and $k \in \mathbb{N}$, we define

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!}.$$

- (a) Show that we have the following Maclaurin series

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$$

- (b) Show that this recovers the usual binomial formula when $\alpha \in \mathbb{N}$.
- (c) By substituting $-x$ for x above, show that we have the Maclaurin series

$$\sqrt{1-x} = 1 - \sum_{n=1}^{\infty} \frac{1}{n2^{2n-1}} \binom{2n-2}{n-1} x^n.$$

- (d) For any positive integer d , we have the following Maclaurin series converging for $|x| < 1$,

$$(1-x)^{-d} = \sum_{n=0}^{\infty} \binom{d+n-1}{n} x^n.$$

7. Find the Maclaurin series for $\tan^{-1} x$

Partial answers, comments and hints

1. Just use term-by-term differentiation on $A(x)$.
2. $e^{2x} = \sum_{n=0}^{\infty} \frac{2^n x^n}{n!} = \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} \right)^2$ where the Cauchy product on the right can be computed using the binomial formula $\sum_{j=0}^n \binom{n}{j} = (1+1)^n = 2^n$.
3. $\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$, $\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$. Note that only even powers of x occur in the Maclaurin series for $\cos x$ whilst only odd ones occur for $\sin x$.
4. This should be straightforward, just make sure you check the constant of integration.

5.

$$\ln(1-x) = \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1} = \sum_{n=1}^{\infty} \frac{1}{n} x^n, \text{ for } |x| < 1.$$

6. When $\alpha \in \mathbb{N}$ we have $\binom{\alpha}{k} = 0$ whenever $k > \alpha$. For $\alpha = \frac{1}{2}$,

$$\begin{aligned} \binom{1/2}{n} &= \frac{1}{n!} \frac{1}{2} \frac{-1}{2} \frac{-3}{2} \cdots \frac{3-2n}{2} = \frac{(-1)^{n-1}}{n! 2^n} \frac{(2n-2)!}{2 \times 4 \times 6 \times \cdots \times (2n-2)} \\ &= \frac{(-1)^{n-1}}{n! 2^{2n-1}} \frac{(2n-2)!}{(n-1)!} = \frac{(-1)^{n-1}}{n 2^{2n-1}} \binom{2n-2}{n-1} \end{aligned}$$

When α is a negative integer, an easier application of the same trick works.

7. Integrate term-by-term the Maclaurin series $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots$ to get

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

Interestingly, this holds even for $x = 1$ giving $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \dots$, though it converges too slowly to use to approximate π efficiently.

Extension III Mathematics Activity Sheet

Counting Monomials

Before doing this Activity Sheet, do watch the teaser video on the web-page above. The goal is the following

Formula. For any $n \in \mathbb{N}$, the number u_n of solutions to $d_1 + \dots + d_r = n$ where $d_1, \dots, d_r \in \mathbb{N}$ is $\binom{r+n-1}{n}$.

We will prove this formula a couple of ways in this Activity Sheet. One very enlightening way, involves considering variables $x_1, x_2, \dots, x_r$. The solutions above then correspond to *monic monomials*, that is, polynomials of the form $m(x_1, \dots, x_r) = x_1^{d_1} x_2^{d_2} \dots x_r^{d_r}$ where $m(x, \dots, x) = x^n$, that is, has *total degree* n .

To do this Activity Sheet, it is good to have some familiarity with the binomial series which can be found on the Activity Sheet Power Series.

Activities

1. In this activity, we prove the formula above by counting monomials and using generating functions. Let

$$P(x_1, \dots, x_r) = \frac{1}{1-x_1} \frac{1}{1-x_2} \dots \frac{1}{1-x_r} \quad (1)$$

$$= (1 + x_1 + x_1^2 + \dots) \dots (1 + x_r + \dots)$$

- (a) Show that product of the power series in (1) is a sum of every monic monomial exactly once.
 - (b) Show that $\frac{1}{(1-x)^r} = \sum_{n=0}^{\infty} u_n x^n$.
 - (c) Hence prove the formula above by using the binomial series to expand $\frac{1}{(1-x)^r}$.
2. Actually, there is a much easier and more elementary way to prove the formula as follows. For each monomial $x_1^{d_1} x_2^{d_2} \dots x_r^{d_r}$, we associate a string of d_1 x 's, followed by a separator $|$, then another d_2 x 's, followed by another separator $|$ and so forth. Thus for example,

$$x_1^3 x_2 x_3^2 x_5 \mapsto xxx|x|xx|x$$

Note there are $r - 1$ separators, so the number of strings on the right with n x 's is $\binom{n+r-1}{n}$. This gives the formula. Work through this argument properly.

3. Find the number of non-negative integer solutions to the equation $d_1 + d_2 + \dots + d_r = 10$.
4. Find the number of positive integer solutions to $d_1 + d_2 + \dots + d_5 = 20$. Hint: use a change of variables for the d_i and the formula.
5. Find the number of non-negative integer solutions to $d_1 + d_2 + \dots + d_5 = 45$ where all $d_i \leq 10$. Hint: use another change variables and the formula.
6. Make up some of your own questions, and see which ones you can answer with the change of variable methods above and which ones you can't.
7. Find the number of non-negative integer solutions to the equation $d_1 + d_2 + \dots + d_r = 10$ such that d_1 is even. Hint: Change P in Q1 above so that only even powers of x_1 occur in the monomials.
8. Can you vary the question above?

Partial answers, comments and hints

1. Part a) follows by expanding, each monomial in the product corresponds to a product of a monomial term in each factor $(1+x_i+\dots)$. For part b), just let all the $x_i = x$ and note that a monomial term $x_1^{d_1} x_2^{d_2} \dots x_r^{d_r}$ then contributes 1 to the co-efficient of x^n precisely when $d_1 + \dots + d_r = n$.
- 2.
3. Just apply the formula to get $\binom{r+9}{10}$.
4. Use the change of variables $c_i = d_i - 1$ so we get the equivalent equation $c_1 + \dots + c_5 = (d_1 - 1) + \dots + (d_5 - 1) = 20 - 5 = 15$. The constraint on the c_i 's is merely that they are in $\mathbb{N}$ now so the answer by the formula is $\binom{19}{15}$.

5. Use the change of variables $c_i = 10 - d_i$ so the new equation is

$$c_1 + \dots + c_5 = (10 - d_1) + \dots + (10 - d_5) = 50 - 45 = 5. \quad (2)$$

The condition $0 \leq d_i \leq 10$ now becomes $0 \leq c_i \leq 10$. However, if all the $c_i \geq 0$ and equation (2) holds, then the condition $c_i \leq 10$ is automatic. We may thus use the formula to get $\binom{9}{5}$.

6. The method of change of variables is basically saying that it gives a bijection (see Activity Sheet Counting and Bijections if you don't know what this means) between the set of solutions to the equation in the formula, to another set of solutions to a new equation.

7. To ensure you only use monomials with even powers of x_1 , use

$$\begin{aligned} Q(x_1, \dots, x_r) &= (1 + x_1^2 + x_1^4 + \dots)(1 + x_2 + x_2^2 \dots) \dots (1 + x_r + \dots) \\ &= \frac{1}{1 - x_1^2} \frac{1}{1 - x_2} \dots \frac{1}{1 - x_r} \end{aligned} \quad (3)$$

The generating function $U(x)$ for the number of solutions is now

$$Q(x, \dots, x) = \frac{1}{(1+x)(1-x)^r} = \frac{2^{-r}}{1+x} + \frac{2^{-r}}{1-x} \dots + \frac{2^{-1}}{(1-x)^r} \quad (4)$$

from which the number of solutions can be read off using the binomial series formula. Perhaps the easiest way to get the partial fraction expansion above is to use the Heaviside cover-up method to find the first term (i.e. numerator above $1+x$) and then divide

$$1 - \frac{(1-x)^r}{2^r} = \frac{1+x}{2} \left(1 + \frac{1-x}{2} + \dots + \left(\frac{1-x}{2} \right)^{r-1} \right)$$

by $(1+x)(1-x)^r$.

8. You can replace the x_i 's in the polynomial P of Q1, with other integer powers of the x_i 's to create a host of new examples.

Extension III Mathematics Activity Sheet Formal Power Series

Sometimes (for example when working with generating functions), it is convenient to allow power series such as $\sum_{n=0}^{\infty} n!x^n$ which do not converge (except at $x = 0$). To deal with this we introduce *Laurent series in x* which are formal expressions of the form $\sum_{n=M}^{\infty} a_n x^n$ where $M \in \mathbb{N}, a_n \in \mathbb{C}$. It is often convenient to let $a_n = 0$ for all $n < M$. The set of all Laurent series is denoted $\mathbb{C}((x))$. The subset of those Laurent series where we can take $M = 0$ above is denoted $\mathbb{C}[[x]]$ and the elements are called *formal power series*. The word “formal” is meant to encapsulate the idea that x is just a symbol, there is no intention of substituting any numbers for x . If some co-efficients a_n are 0, then we can omit the whole term from the expression so for example we have a power series $5 + x = 5 + x + 0x^2 + 0x^3 + \dots$

One can manipulate Laurent series algebraically in much the same way as we can convergent power series. In particular, we have the following operations for Laurent series $A(x) = \sum_{n=M}^{\infty} a_n x^n$, $B(x) = \sum_{n=N}^{\infty} b_n x^n$.

- Addition: $A(x) + B(x) = \sum_{n=\min\{M,N\}}^{\infty} (a_n + b_n)x^n$
- Multiplication:

$$A(x)B(x) = \sum_{n=M+N}^{\infty} \left(\sum_{j=0}^n a_j b_{n-j} \right) x^n$$

- Differentiation: $\frac{dA}{dx}(x) = A'(x) := \sum_{n=M-1}^{\infty} n a_n x^{n-1}$.
- Although one cannot substitute arbitrary numbers for x , it does make sense to define $A(0) = a_0$ if $A(x) \in \mathbb{C}[[x]]$.

One of the main aims of this Activity Sheet is to see that these operations behave much as they do for numbers, in mathematical jargon, we say that $\mathbb{C}((x))$ forms a *field*, as defined below. This should be enough to make all the counting arguments via generating functions legal.

Logically, you need not have completed the Activity Sheet on Power Series to do this one, but it might make more sense to do so.

Activities

1. Consider the formal power series $A(x) = 1 + x + x^2 + \dots$. Show that $(1 - x)A(x) = 1$. Hence we will often write $(1 - x)^{-1}$ or $\frac{1}{1-x}$ as shorthand for $A(x)$.
2. Let $A(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. Show that $A'(x) = A(x)$ and $A(0) = 1$ so it makes sense to use the notation $e^x = A(x)$, something we will do.
3. Show that the multiplication is *commutative*, that is, $A(x)B(x) = B(x)A(x)$.
4. If we have another Laurent series $C(x) = \sum_n c_n x^n$, show that the *distributive law* holds, that is, $A(x)(B(x) + C(x)) = A(x)B(x) + A(x)C(x)$.
5. Show that multiplication is *associative*, that is, $(A(x)B(x))C(x) = A(x)(B(x)C(x))$ so we can define these equal Laurent series to be the triple product written $A(x)B(x)C(x)$ without brackets. More generally, one can multiply any finite number of Laurent series without specifying brackets.
6. Let A be a set. A *binary operation*, say denoted $+$, on A is a rule assigning to any ordered pair of elements $a, b \in A$, a new element $a + b$ in A . That is, it is a function $A \times A \rightarrow A$. Such a set equipped with a binary operation is an *abelian group*, if it satisfies the following axioms:
 - $+$ is associative and commutative, that is $(a + b) + c = a + (b + c)$, $a + b = b + a$ for all $a, b, c \in A$.
 - Existence of Zero: There is an element denoted $0 \in A$ and called *zero*, such that $0 + a = a$ for all $a \in A$.
 - Existence of negatives: For each $a \in A$, there is an element denoted $-a$ and called the *negative of a*, such that $a + (-a) = 0$.

Show that $\mathbb{C}((x))$ is an abelian group when equipped with addition for the binary operation $+$.

7. A subset B of an abelian group A (with addition $+$), is a *subgroup* if it satisfies the following *subgroup closure axioms*:
 - $0 \in B$

- (closure under addition) If $b, b' \in B$ then $b + b' \in B$
- (closure under negatives) If $b \in B$ then $-b \in B$.

In this case, show that $+$ restricts to a binary operation on B and B itself is an abelian group. Show that $\mathbb{C}[[x]]$ is a subgroup of $\mathbb{C}((x))$ so is an abelian group.

8. A *ring* is an abelian group R with addition say $+$, equipped with a second binary operation, called *ring multiplication* which sends an ordered pair $a, b \in R$ to $ab \in R$, which satisfies the following *ring axioms*:
 - Existence of 1: There is an element denoted $1 \in R$ and called *one* or the *multiplicative identity*, such that $1a = a1 = a$ for all $a \in R$.
 - Multiplication is associative
 - The distributive laws hold: For any $a, b, c \in R$ we have both $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

We say R is *commutative* if ring multiplication is commutative. Show that $\mathbb{C}((x))$ is a commutative ring.

9. Show that $\mathbb{C}[[x]]$ is a commutative ring when equipped with addition and multiplication of Laurent series for ring addition and multiplication.
10. In a ring R , the *inverse* of an element $r \in R$ is an element, usually denoted r^{-1} or $\frac{1}{r}$, which satisfies $rr^{-1} = r^{-1}r = 1$ (see Q1 for an example). Consider a power series $A(x) = \sum_{n=0}^{\infty} a_n x^n$ with $a_0 \neq 0$. Show that $A(x)$ has an inverse $\sum_{n=0}^{\infty} b_n x^n$ by solving for the coefficients b_n inductively.
11. A *field* is a commutative ring $\mathbb{F}$ where every non-zero element α is *invertible*, that is, has an inverse. Show that $\mathbb{C}((x))$ is a field.
12. The point of having a field, is that it is an environment where you can perform many of the algebraic manipulations you are accustomed to for numbers. For example, we would like to do partial fraction expansions of power series. This holds since we have the following formula which holds in any field $\mathbb{F}$

Formula $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$, for $a, b, c, d \in \mathbb{F}, c, d \neq 0$ where fractions like $\frac{a}{b}$ are interpreted as ab^{-1} (which $= b^{-1}a$ by commutativity of multiplication!).

Prove this formula as follows

- (a) Show that the following cancellation law holds in a field: given $u, v, w \in \mathbb{F}$ with $w \neq 0$, then $uw = vw \implies u = v$.
 - (b) It thus suffices to show that $bd\left(\frac{a}{b} + \frac{c}{d}\right) = ad + bc$. (Note that we have already used the axiom of associative multiplication to write the LHS unambiguously here).
13. Suppose a, b are invertible elements of a ring R . Show that ab is invertible and give an expression for its inverse.
 14. You have seen many standard operations on functions (such as sums, products, derivatives) can be carried out on formal power series. Can you come up with any others?

Partial answers, comments and hints

1. Just use the definition of the (Cauchy) product above.
- 2.
3. Either change the summation variable j in the definition of multiplication above to $n - j$, or, more symmetrically, show that the product can be defined by the formula

$$A(x)B(x) = \sum_{n=M+N}^{\infty} \left(\sum_{i+j=n} a_j b_i \right) x^n$$

4.

$$\begin{aligned} \text{LHS} &= \sum_n \left(\sum_j a_j (b_{n-j} + c_{n-j}) \right) x^n \\ &= \sum_n \left(\sum_j a_j b_{n-j} + \sum_j a_j c_{n-j} \right) x^n = \text{RHS}. \end{aligned}$$

5. The best way to do this is to show both triple products can be written

$$A(x)B(x)C(x) = \sum_n \left(\sum_{i+j+k=n} a_i b_j c_k \right) x^n$$

6. The binary operation of an abelian group is often called *addition*, especially if one uses $+$ to denote it. One should note that the zero and negative in an abelian group are unique in the sense that the elements $0, -a$ are uniquely determined by their defining property. Note that this means we can subtract by defining $a - b := a + (-b)$.

In the case of $\mathbb{C}((x))$, the zero is $0 = 0 + 0x + 0x^2 + \dots$ and the negative of $A(x) = \sum_n a_n x^n$ is $-A(x) = \sum_n (-a_n) x^n$. The axioms for associativity and commutativity for addition of Laurent series follow by applying the associative and commutative laws for complex numbers coefficientwise. Make sure you know what this means and how to write out the proper proofs.

7. To see that B is an abelian group, note that closure under addition means that restricting inputs to B , the addition in A restricts to a binary operation in B which has the same zero and where existence of negatives hold (with same negative) by closure under negatives. Associativity and commutativity of $+$ in B follows from that in A . The subgroup closure axioms for $\mathbb{C}[[x]]$ should be easy to check since if you add or negate Laurent series where the coefficients of negative powers of x are 0, then those coefficients stay 0.
8. The multiplicative identity of $\mathbb{C}((x))$ is $1 = 1 + 0x + 0x^2 + \dots$. In general, the multiplicative identity of ring is uniquely determined by its defining property.
9. The smart way to do this, is to realise that not only is $\mathbb{C}[[x]]$ a subgroup of $\mathbb{C}((x))$, it is also a *subring* in the sense that we also have $1 \in \mathbb{C}[[x]]$ and *closure under multiplication*, that is, the product of two power series is again a power series. Then all the axioms for a commutative ring are inherited from those of $\mathbb{C}((x))$.
10. We wish to show the following result by induction on k . There exist $b_0, \dots, b_k \in \mathbb{C}$ such that for any choice of $b_{k+1}, b_{k+2}, \dots$, letting $B(x) = \sum_n b_n x^n$, we have the product $A(x)B(x)$ has the form $1 + c_{k+1}x^{k+1} + c_{k+2}x^{k+2} + \dots$. Indeed, the base case is given by letting $b_0 = a_0^{-1}$. Suppose the result holds for $k-1, k \geq 1$. We define b_k to be the unique solution to

$$a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0 = 0.$$

so the coefficient of x^k in the product $A(x)B(x)$ is 0.

11. If the Laurent series $A(x) \neq 0$, then we can write $A(x) = x^M B(x)$ where $B(x)$ is a power series with $B(0) \neq 0$. Thus $B(x)^{-1}$ exists and $A(x)^{-1} = x^{-M} B(x)^{-1}$.
12. For the cancellation law, note that w has an inverse w^{-1} so by field axioms (and in particular omitting brackets around triple products) we have

$$u = u1 = uww^{-1} = vww^{-1} = v1 = v.$$

Part b) is proved similarly using axioms step-by-step. We then apply the cancellation law to $bd \left(\frac{a}{b} + \frac{c}{d} \right) = (ad + bc)(bd)^{-1}(bd)$. There are other ways to prove the formula, but the cancellation law is good to know.

13. $(ab)^{-1} = b^{-1}a^{-1}$ since

$$b^{-1}a^{-1}(ab) = b^{-1}1b = b^{-1}b = 1$$

and similarly $(ab)b^{-1}a^{-1} = 1$.

14. Another operation that we have already used is substitutions of the form $x \mapsto 2x$ and so forth. More generally, you might wish to show that if $F(x), G(x) \in \mathbb{C}[[x]]$ are such that $G(0) \neq 0$, then it makes sense to define $F(G(x))$. It is a good exercise to think through how this works.

Extension III Mathematics Activity Sheet Catalan Numbers

Catalan numbers appear in many combinatorial counting problems. For example, the number u_n of ways you can bracket the product $x_1x_2\ldots x_n$ is the $n - 1$ -st Catalan number. This example is looked at in depth in the associated video which you can find in the webpage above. There we show the Catalan numbers satisfy the recurrence relation

$$u_n = \sum_{j=0}^{n-1} u_j u_{n-1-j}, \quad \text{for } n \geq 2 \quad (1)$$

if we define the degenerate value $u_0 = 0$. If $U(x) = \sum_n u_n x^n$ is the corresponding generating function, then this gives the relation $U - U^2 = x$. Solving this quadratic and using the binomial series gives $u_n = \frac{1}{n} \binom{2n-2}{n-1}$.

In this Activity Sheet, we look at some other combinatorial problems which give rise to Catalan numbers. The approach we will take to studying these problems is generating functions so ideally, you should be familiar with the Activity Sheets on (Formal) Power Series and Generating Functions, before tackling this one.

Activities

1. A *2n-term Dyck sequence* is a sequence $a_1, a_2, \ldots, a_{2n}$ of n 1's and n -1's, ordered in such a way that the partial sums $\sum_{i=1}^m a_i \geq 0$ for all m . For example, $1, -1, -1, 1$ is not a Dyck sequence, but $1, -1, 1, -1$ is. Let c_n be the number of $2n$ -term Dyck sequences. By default, we set $c_0 = 1$.
 - (a) Show that c_n satisfies the recurrence relation $c_{n+1} = \sum_{j=0}^n c_j c_{n-j}$ for $n \geq 1$.
 - (b) Show that the generating function $C(x) = \sum_{n=0}^{\infty} c_n x^n$ satisfies $xC(x)^2 = C(x) - 1$.
 - (c) Hence show $xC(x) = U(x)$ and so c_n is the n -th Catalan number.
2. Let P_n be a convex n -gon. Let t_n be the number of ways that you can draw $n - 3$ non-crossing diagonals of P_n so as to divide it into

triangles. By default, we let $t_n = 0$ for $n < 2$. Show that t_{n+2} is also the n -th Catalan number.

3. Look up some other combinatorial problems that give Catalan numbers and see if you can show they indeed give Catalan numbers by using the techniques here.

Partial answers, comments and hints

1. Suppose given a $2n + 2$ -term Dyck sequence $a_1, \dots, a_{2n+2}$. Consider the first time the partial sum $\sum_{i=1}^l a_i$ is zero, which at the very latest is $l = 2n + 2$. Then l is even, say $l = 2j + 2$ and we must have $a_1 = 1, a_{2j+2} = -1$. Hence $a_2, \dots, a_{2j+1}$ is a $2j$ -term Dyck sequence. It can be quite arbitrary so the number of possibilities for this is c_j . The possibilities for $a_{2j+3}, \dots, a_{2n+2}$ can be chosen independently to be any $2(n - j)$ -term Dyck sequence, so the recurrence relation follows.

$$xC(x)^2 = x \sum_{n=0}^{\infty} \left(\sum_{j=0}^n c_j c_{n-j} \right) x^n = \sum_{n=0}^{\infty} c_{n+1} x^{n+1} = C(x) - c_0 = C(x) - 1$$

Thus $V(x) := xC(x)$ thus satisfies the equation

$$V(x)^2 = x^2 C(x)^2 = x(C(x) - 1) = V(x) - x$$

which is the same quadratic that $U(x)$ satisfies. However, the other root of the quadratic involves negative coefficients, so cannot be $V(x)$ either. Thus $V(x) = xC(x) = U(x)$.

2. Suppose we label the vertices of P_n clockwise by $v_1, \dots, v_n$. If we break up P_n into triangles by drawing $n - 3$ diagonals, then the side $v_n v_1$ is also the side of one of those triangles, say one whose third vertex is v_i . Remove $\triangle v_1 v_i v_n$ from P_n to get a convex i -gon and convex $(n + 1 - i)$ -gon which are each divided into triangles by the remaining $n - 5$ diagonals. This gives the relation $t_n = \sum_{i=2}^{n-1} t_i t_{n+1-i}$. You can now play our usual game, or shift indices (that is define $u_n = t_{n+1}$) to give one of our old recurrence relations.

Extension III Mathematics Activity Sheet Counting and Bijections

You might wish to watch the associated video on the webpage above before starting this Activity Sheet. We've summarised though, the main points below.

Formally, the notion of counting and associated theory is encoded in the mathematical notion of bijections as follows. For $n \in \mathbb{N}$, let $[n] = \{1, \dots, n\}$ so, in particular, $[0] = \emptyset$.

Definition 0.1 Consider a function $f: A \rightarrow B$ and recall the pre-image of $b \in B$ is $f^{-1}(b) := \{a \in A \mid f(a) = b\}$. We say f is

1. injective or 1-1 if $f^{-1}(b)$ has at most 1 element for every $b \in B$,
2. surjective or onto if $f^{-1}(b) \neq \emptyset$ for every $b \in B$,
3. bijective if f is both 1-1 and onto, in which case we get a function $f^{-1}: B \rightarrow A$ where $f^{-1}(b)$ is the unique solution a to $f(a) = b$.

We say that a set A has cardinality $|A| = n$ if there's a bijection $[n] \rightarrow A$. It is intuitively obvious that the cardinality is well-defined (that is, you can't get bijections with both $[n]$ and $[m]$ unless $n = m$) and that bijective sets have the same cardinality. We prove this formally in this Activity Sheet, along with other interesting facts associated with bijections.

Activities

1. Show that $f: \mathbb{Z} \rightarrow \mathbb{N} : n \mapsto n^2$ is neither injective nor surjective but that $g: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0} : \alpha \mapsto \alpha^2$ is bijective. Note $\mathbb{R}_{\geq 0} = \{\alpha \in \mathbb{R} \mid \alpha \geq 0\}$.
2. If A is a set, the *identity on A* is the function $\text{id}_A: A \rightarrow A: a \mapsto a$. Is id_A injective? surjective? bijective?
3. Recall that given functions $f: A \rightarrow B$, $g: B \rightarrow C$, the composite function denoted gf or $g \circ f: A \rightarrow C$ is defined by $(gf)(a) = g(f(a))$. Show that $f \text{id}_B = f = f \text{id}_A$. Show also that function composition is associative, if we have another function $h: C \rightarrow D$, then $(hg)f = h(gf)$.

4. Show that the composite of injective (respectively, surjective, bijective) functions is injective (respectively surjective, bijective).
5. Suppose $f: A \rightarrow B$ is a bijection and A has cardinality n . Prove that B also has cardinality n .
6. Let $f: A \rightarrow B$ be bijective. Show that $f^{-1}: B \rightarrow A$ satisfies $ff^{-1} = \text{id}_B$, $f^{-1}f = \text{id}_A$. Show also that f^{-1} is bijective.
7. Let $f: A \rightarrow B$ be a function. We say it is *invertible* if there is a function $g: B \rightarrow A$ such that $gf = \text{id}_A$, $fg = \text{id}_B$. Show that in this case we have f is a bijection and $g = f^{-1}$.
8. Prove by induction on n , that if $f: [n] \rightarrow [n]$ is injective, then it is also surjective and hence bijective. (This is hard).
9. Show that cardinality is well-defined in the sense that if there are two bijections $f: [n] \rightarrow A$, $g: [m] \rightarrow A$, then $n = m$. Such sets are said to be *finite*. Hint: we may assume $n \leq m$ and use the previous question on $f^{-1}g$.
10. A *permutation* of a set A is a bijection of the form $\sigma: A \rightarrow A$. The set of these is (here) denoted $\text{Perm}(A)$.
 - (a) Find the cardinality of $\text{Perm}([n])$.
 - (b) Let $f: A \rightarrow B$ be a bijection. Show that $\Phi: \text{Perm}(A) \rightarrow \text{Perm}(B): \sigma \mapsto f\sigma f^{-1}$ is a well-defined bijection.

Partial answers, comments and hints

1. Note that $f^{-1}(4) = \{-2, 2\}$ so f is not injective whilst $f^{-1}(3) = \emptyset$ so neither is it surjective. On the other hand, given any $\beta \in \mathbb{R}_{\geq 0}$, there is a unique solution $\alpha = \sqrt{\beta}$ to $g(\alpha) = \beta$ so g is bijective and $g^{-1} = \sqrt{}$.
2. id_A is bijective.
3. To show two functions (with the same domain and codomain are the same), we need to show they have the same function values for every possible input. For example, for any $a \in A$,

$$(\text{id}_B f)(a) = \text{id}_B(f(a)) = f(a)$$

so $\text{id}_B f = f$. Similarly one proves $f = f \text{id}_A$ and associativity of composition.

4. This is a standard fact you can find in many standard texts in calculus or discrete maths. We prove that if f, g is injective, so is gf . Let $c \in C$ and $a \in (gf)^{-1}(c)$. Then by definition of pre-image $c = (gf)(a) = g(f(a))$. Hence $b := f(a) \in g^{-1}(c)$ and is in fact the only element of $g^{-1}(c)$ since g is injective. We have thus shown that, every element of $(gf)^{-1}(c)$ lies in $f^{-1}(b)$. But f is injective, so $f^{-1}(b)$ consists of the single element a and we are done.
5. Since A has cardinality n , there is a bijection $h: [n] \rightarrow A$. Now the composite $fh: [n] \rightarrow B$ is also a bijection by Q4 so B also has cardinality n .
6. For any $b \in B$, $f^{-1}(b)$ is the unique $a \in A$ such that $f(a) = b$. Thus $(ff^{-1})(b) = f(f^{-1}(b)) = f(a) = b$ so $ff^{-1} = \text{id}_B$, the two functions having the same function values for all possible inputs $b \in B$. Conversely, given $a \in A$ let $b = f(a)$ so again we have $f^{-1}(b) = a$. Then $(f^{-1}f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a$ so $f^{-1}f = \text{id}_A$.

We now show that f^{-1} is bijective. Let $a \in A$ so we wish to show there is a unique solution $b \in B$ to the equation $f^{-1}(b) = a$. Note that $b = f(a)$ is a solution since on substituting we find $f^{-1}(b) = f^{-1}f(a) = \text{id}_A(a) = a$ whilst it is also the only possible solution since applying f to the $f^{-1}(b) = a$ yields $f(a) = ff^{-1}(b) = \text{id}_B(b) = b$.

7. This is a standard result that you can find in any discrete maths text or most calculus texts to. Just repeat the proof in the previous question and show it works more generally here.
8. The base cases $n = 0, 1$ hold since the only functions $[0] \rightarrow [0]$ and $[1] \rightarrow [1]$ are the identity which is bijective by Q 2. Suppose the result holds true in the case $n - 1$ and consider an injective function $f: [n] \rightarrow [n]$. Then $f^{-1}(n)$ has at most one element so there are two cases depending on whether this set is empty or not. In the former case, we have $f(i) \in [n - 1]$ for all $i \in [n]$ so we can restrict the domain of f to $[n - 1]$ to get a new function

$g := f|_{[n-1]}: [n-1] \rightarrow [n-1]: i \mapsto f(i)$. Note that g is also injective since for any $j \in [n-1]$ we have $g^{-1}(j) = f^{-1}(j)$. We may thus apply the inductive hypothesis to g , to see that g is surjective. Thus there is $i \in [n-1]$ such that $f(i) = g(i) = f(n)$. This contradicts the injectivity of f since then $f^{-1}(f(n))$ contains at least n, i .

We must thus be in the second case where $f^{-1}(n)$ is non-empty so is the single element set $\{i\}$. If $i = n$, then any $j \in [n-1]$ is not in $f^{-1}(n)$ so $f(j) \in [n-1]$. We thus get a restricted map $g := f|_{[n-1]}: [n-1] \rightarrow [n-1]: j \mapsto f(j)$ which is surjective by the inductive hypothesis. Thus it is also true that $f^{-1}(j) = g^{-1}(j)$ contains a unique element for all $j \in [n-1]$ as well so f is bijective.

The remaining case to prove is where $i \in [n-1]$. Note that $f(n) \in [n-1]$ for otherwise we have $n \in f^{-1}(n) = \{i\}$. We then apply the inductive hypothesis to the function $g: [n-1] \rightarrow [n-1]$ defined by $g(j) = f(j)$ if $j \neq i$ and $g(i) = f(n)$. This is possible because injectivity of g is seen from computing $g^{-1}(f(n)) = \{i\}$ and $g^{-1}(j) = f^{-1}(j)$ when $j \neq f(n)$. Thus we have non-empty $f^{-1}(j) = g^{-1}(j)$ for $j \neq f(n)$ or n and non-empty $f^{-1}(f(n)) \ni n$ and $f^{-1}(n) = \{i\}$. Thus f is surjective.

9. Note that f^{-1} is bijective by Q 6 so $f^{-1}g: [m] \rightarrow [n]$ is bijective by Q 4. Now $n \leq m$ so $[n]$ is a subset of $[m]$ and we may consider the function $h: [n] \rightarrow [n]: i \mapsto f^{-1}g(i)$. Note that $h^{-1}(j)$ is empty if $j > n$ and is the single element set $(f^{-1}g)^{-1}(j)$ otherwise. Thus h is also injective. Now Q 8, shows that h is surjective so $[n] = [m]$ and thus $m = n$.
10. $|\text{Perm}([n])| = n!$. Note that $f\sigma f^{-1}$ has domain and codomain B and is furthermore a bijection by Q4,6. Thus Φ is well-defined. Note that $\Psi: \text{Perm}(B) \rightarrow \text{Perm}(A): \sigma \mapsto f^{-1}\sigma f$ is similarly a well-defined map. It remains only to check $\Psi\Phi = \text{id}$, $\Phi\Psi = \text{id}$ which is easy.

Extension III Mathematics Activity Sheet

Linear Ordinary Differential Equations

Watch the associated video on “Linear Ordinary Differential Equations” on the webpage above and then try your hand at the following Activity Sheet.

A *first order linear ordinary differential equation* (1st order ODE for short) is one of the form

$$\frac{dy}{dx} + g(x)y = h(x) \quad (1)$$

where the functions $g(x), h(x)$ are given, and one wishes to solve for the function $y = f(x)$. There is a standard procedure for solving it, where you multiply the equation by a function $\rho(x)$, called an *integrating factor*, especially chosen so that $\frac{d\rho}{dx} = \rho(x)g(x)$. This has the effect that the change of variables $z = \rho(x)y$ simplifies the ODE (1) to

$$\frac{dz}{dx} = \rho(x)\frac{dy}{dx} + \frac{d\rho}{dx}y = \rho(x)\frac{dy}{dx} + \rho(x)g(x)y = \rho(x)h(x) \quad (2)$$

One simply integrates (2) to solve for z , from which y can be obtained as $y = \frac{z}{\rho(x)}$.

To do this Activity Sheet, you only need to know basic high school calculus and complex numbers.

Activities

1. Solve the linear ODE $\frac{dy}{dx} + \frac{y}{x} = 4x^2$
2. Find the function $y = f(x)$ such that

$$\frac{dy}{dx} + y = e^{2x}$$

and satisfying the *initial condition* $y = 1$ when $x = 0$, that is, $f(0) = 1$. Hint: you will need to pick the integration constant so that the initial condition holds.

3. Consider the *second order* differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0 \quad (3)$$

Order 2 refers to the second derivative involved. We will reduce it to a pair of *cascaded* first order ones as follows.

- (a) Consider the new variable $u = \frac{dy}{dx} - 2y$. Show that u satisfies the first order equation $\frac{du}{dx} - 3u = 0$ and solve it.
 - (b) Using your solution in the previous part, solve $u = \frac{dy}{dx} - 2y$ for y .
 - (c) Find the solution with initial conditions $y(0) = 2, y'(0) = 5$.
 - (d) Show this trick also works with using the variable $v = \frac{dy}{dx} - 3y$.
4. Here we solve the second order linear ODE

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$$

using the change of variable $u = \frac{dy}{dx} + x$.

5. What second order ODEs can you solve using the technique above?
6. Let $f(x) = g(x) + ih(x)$ be a complex-valued function of a real variable x (here g, h are real-valued). Then we can define

$$\begin{aligned} f'(x) &= \lim_{\delta \rightarrow 0} \frac{f(x+\delta) - f(x)}{\delta} \\ &= \lim_{\delta \rightarrow 0} \frac{g(x+\delta) - g(x)}{\delta} + i \lim_{\delta \rightarrow 0} \frac{h(x+\delta) - h(x)}{\delta} = g'(x) + ih'(x). \end{aligned} \quad (4)$$

- (a) Show that the product rule still holds.
 - (b) Recall $e^{x+iy} = e^x(\cos x + i \sin y)$. Show that for any $a \in \mathbb{C}$ we have $\frac{d}{dx}(e^{ax}) = ae^{ax}$.
7. Consider the ODE $\frac{d^2y}{dx^2} + y = 0$. Show that $y = \sin x$ is a solution, and find the general solution.

Partial answers, comments and hints

1. Use the change of variables $z = xy$ so $\frac{dz}{dx} = x\frac{dy}{dx} + y = 4x^3$. Hence $z = x^4 + C$ where C is an arbitrary integration constant. Thus $y = x^{-1}z = x^3 + Cx^{-1}$. Note that the solution is not unique, one usually imposes an *initial condition* as in the next question to get a single function.
2. Use change of variables $z = e^x y$ so $\frac{dz}{dx} = e^x \frac{dy}{dx} + e^x y = e^{3x}$. Thus $z = \frac{1}{3}e^{3x} + C$, for some integration constant C . Hence $y = e^{-x}y = \frac{1}{3}e^{2x} + Ce^{-x}$. We now substitute $x = 0, y = 1$ to find $1 = \frac{1}{3} + C$ so $C = \frac{2}{3}$ giving the final answer as $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$.
3. $\frac{du}{dx} - 3u = \frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3(\frac{dy}{dx} - 2y) = 0$. Hence $u = Ce^{3x}$ for some constant C . We now solve $\frac{dy}{dx} - 2y = Ce^{3x}$. The integrating factor is e^{-2x} and we see $\frac{d}{dx}(e^{-2x}y) = e^{-2x}\frac{dy}{dx} - 2e^{-2x}y = Ce^x + D$ for some constant D . Thus $y = Ce^{3x} + De^{2x}$. Note second order means there are now two integration constants involved. Note $y(0) = C + D, y'(0) = 3C + 2D$. Solving simultaneously gives $C = D = 1$. For v , note that $\frac{dv}{dx} - 2v = 0$.
4. Now $\frac{du}{dx} + u = 0$ so $u = Ce^{-x}$ for some constant C . Solve $\frac{dy}{dx} + y = Ce^{-x}$ with the integrating factor e^x to see $\frac{d}{dx}(e^x y) = e^x \frac{dy}{dx} + e^x y = C$. Hence $y = Cxe^x + De^x$.
5. Suppose there are real numbers α, β such that $\frac{du}{dx} - \alpha u = 0$ and $\frac{dy}{dx} - \beta y = 0$. Then on substituting we find

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0 \quad (5)$$

where $a = -\alpha - \beta, b = \alpha\beta$. Thus we can certainly solve (5) whenever the quadratic $z^2 + az + b$ has real roots. Actually, as seen in the next two question, the method works even for complex α, β so a, b can be arbitrary.

6. These are both long but elementary calculations. For example, if $u_j = g_j(x) + ih_j(x)$ for $j = 1, 2$, then the product rule means

$$\frac{d}{dx}((g_1 + ih_1)(g_2 + ih_2)) = (g'_1 + ih'_1)(g_2 + ih_2) + (g_1 + ih_1)(g'_2 + ih'_2)$$

which follows on expanding both sides.

7. To solve the ODE as a cascaded system, use the variable $u = \frac{dy}{dx} + iy$ which satisfies $\frac{du}{dx} - iu = 0$. The general solution is $y = Ce^{ix} + De^{-ix}$ where the integration constants C, D are now complex. When $C = -\frac{i}{2}, D = \frac{i}{2}$ we get $\sin x$. In general, if you want y to be real-valued, then you better let $D = \bar{C}$ so that Ce^{ix}, D^{-ix} are complex conjugates so sum to a real number. If you let $C = A + iB$, then $Ce^{ix} + \bar{C}e^{-ix} = 2A \cos x - 2B \sin x$. From this one can easily show the general real solution consists of sum of arbitrary real multiples of $\cos x$ and $\sin x$.

Extension III Mathematics Activity Sheet Exponential Generating Functions

Given a sequence of complex numbers $a_0, a_1, \dots$, it is sometimes useful to consider its *exponential generating function*

$$\tilde{A}(x) := \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n \quad (1)$$

as opposed to its usual generating function. For example, suppose $\{b_n\}_{n \in \mathbb{N}}$ is another sequence with exponential generating function $\tilde{B}(x)$. There are many interesting examples where one is interested in a new sequence $\{c_n\}_{n \in \mathbb{N}}$, constructed from these, defined by

$$c_n = \sum_{k=0}^n \binom{n}{k} a_k b_{n-k} \quad (2)$$

If you watch the video “Exponential Generating Functions” in the website above, you will see one such example related to derangements. The exponential generating function $\tilde{C}(x)$ of the sequence $\{c_n\}$ is easily computed from $\tilde{A}(x), \tilde{B}(x)$ since

$$\begin{aligned} \tilde{A}(x)\tilde{B}(x) &= \sum_{k=0}^{\infty} \frac{a_k}{k!} x^k \sum_{l=0}^{\infty} \frac{b_l}{l!} x^l \\ &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{1}{k!(n-k)!} a_k b_{n-k} \right) x^n \\ &= \sum_{n=0}^{\infty} \frac{c_n}{n!} x^n = \tilde{C}(x) \end{aligned} \quad (3)$$

In this Activity Sheet, we will see how exponential generating functions are useful in counting problems and how they are related to other fields of mathematics, such as differential equations. To do it, you should know a little about power series and linear differential equations so you might want to do the Activity Sheets on those topics before trying this one.

Activities

1. Consider the sequence $a_n = 4^n$. Find its exponential generating function $\tilde{A}(x)$. Note: the best answer is as some type of exponential function.
2. Find the exponential generating function $\tilde{A}(x)$ of the sequence $a_n = \binom{n}{3} 7^n$
3. The *left shift* of a sequence $a_0, a_1, \dots$ is the sequence $b_0 = a_1, b_1 = a_2, \dots$ so $b_n = a_{n+1}$ for $n \geq 0$. Show that their exponential generating functions $\tilde{A}(x), \tilde{B}(x)$ are related by $\tilde{B}(x) = \tilde{A}'(x)$.
4. Let $\tilde{A}(x)$ be the exponential generating function for $\{a_n\}_{n \in \mathbb{N}}$. Show that $x\tilde{A}(x)$ is the exponential generating function for the sequence $\{b_n = na_{n-1}\}_{n \in \mathbb{N}}$ (here $b_0 = 0$).
5. It's the festive season, and a group of friends decide that, rather than buying presents for everyone else, they will each buy a present for one other friend, in such a way that every one will receive exactly one present. These correspond to the mathematical notion of *derangements* of a set S , which are permutations σ of S such that $\sigma(s) \neq s$ for all $s \in S$. Let d_n be the number of derangements of $[n]$ and $\tilde{D}(x)$ be the corresponding exponential generating function. We assume the degenerate values $d_0 = 1, d_1 = 0$. In the video, we used Equation (3) to study $\tilde{D}(x)$. Our approach here is via differential equations.
 - (a) Show that $d_{n+1} = nd_n + nd_{n-1}$ for $n \geq 1$. Hint: How many derangements are there of $[n+1]$ which swap $n+1$ with some $i \in [n]$?
 - (b) Show that $(1-x)\tilde{D}'(x) = x\tilde{D}(x)$.
 - (c) Solve the ODE above, noting that the initial condition is given by $\tilde{D}(0) = 1$.
 - (d) Using the formula for $\tilde{D}(x)$ you found and the Cauchy product, find a formula for d_n .
 - (e) If you know about the principle of inclusion-exclusion, give a combinatorial interpretation of the alternating sum you found in the previous part. If you don't know about the principle of inclusion-exclusion, it's not too difficult, you can look it up in any standard discrete maths text. In fact, it's

safe to look up online, since you should be able to tell if the version there is true or not just by thinking through the result logically.

6. This question gives a natural example of a sequence constructed via (2). There's a party of n people. Afterwards, some of them pair up. Let c_n be the number of ways this can happen.
 - (a) Let a_n be the number of ways of splitting up an n -element set into pairs (so n odd in particular means $a_n = 0$). Show that the resulting exponential generating function is $\tilde{A}(x) = e^{\frac{1}{2}x^2}$.
 - (b) Corresponding to those who do not pair up, we consider the constant sequence $b_n = 1$. Show that the relation (2) holds and hence compute the exponential generating function $\tilde{C}(x)$ for $\{c_n\}_{n \in \mathbb{N}}$.
7. We say that a permutation $\sigma \in \text{Perm} S$ permutes the elements of S *cyclically* if we can order all the elements into a sequence $s_1, \dots, s_n$ so that $\sigma(s_i) = s_{i+1}$ for $1 \leq i < n$ and $\sigma(s_n) = s_1$, that is, σ shifts the elements along to the next one in the sequence, and cycles the last one back to the beginning. In this case, we also say that σ has 1 orbit.
 - (a) Let a_n be the number of 1 orbit permutations of $[n]$ and $\tilde{A}(x)$ be its exponential generating function. Show that $\tilde{A}(x) = \log(1 - x)^{-1}$. By default, we set $a_0 = 0, a_1 = 1$ here.
 - (b) We say that $\sigma \in \text{Perm}(S)$ is a *2 orbit* permutation of S if we can partition $S = S_1 \cup S_2$ as a disjoint union of two sets, such that σ permutes the elements of S_1 cyclically, and those of S_2 cyclically. For example, suppose $\sigma \in \text{Perm}(S)$ swaps 1 and 3, and cyclically permutes the others by $2 \mapsto 4 \mapsto 5 \mapsto 2$, then σ has 2 orbits. Let b_n be the number of 2 orbit permutations of $[n]$. Find its exponential generating function using (2).
 - (c) One can similarly define j orbit permutations, where S is a disjoint union of j sets, each of which is cyclically permuted. Find the exponential generating function $\tilde{C}_j(x)$ for the sequence c_n = the number of j orbit permutations of $[n]$.
 - (d) It is an interesting and well-known fact that every permutation is actually a j orbit permutation for some j . We will

give a proof via counting as follows. Note that the number of permutations of $[n]$ is $n!$ so the corresponding exponential generating function is $\tilde{P}(x) = 1 + x + \dots = \frac{1}{1-x}$. Show that $1 + \tilde{A}(x) + \tilde{B}(x) + \tilde{C}_2(x) + \dots = \frac{1}{1-x}$.

Partial answers, comments and hints

1. e^{4x} .
2. $\tilde{A}(x) = \sum_{n=3}^{\infty} \binom{n}{3} \frac{7^n}{n!} x^n = \sum_{n=3}^{\infty} \frac{7^n}{(n-3)!3!} x^n = \sum_{m=0}^{\infty} \frac{7^{m+3}}{m!3!} x^{m+3} = \frac{7^3 x^3}{3!} e^{7x}$ where we have used the change of summation index $m = n - 3$.
3. $\frac{d}{dx} \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n = \sum_{n=0}^{\infty} \frac{na_n}{n!} x^{n-1} = \sum_{n=0}^{\infty} \frac{a_n}{(n-1)!} x^{n-1} = \sum_{m=0}^{\infty} \frac{a_{m+1}}{m!} x^m = \tilde{B}(x)$.
4. This is similar to the previous question.

5. This ODE is best deduced from the previous two questions. Otherwise, you can check directly that the coefficient of x^n in the LHS is $\frac{1}{n!}(d_{n+1} - nd_n)$ whilst that on the RHS is $\frac{1}{n!}nd_{n-1}$.

The ODE is linear: $\tilde{D}'(x) + \frac{x}{x-1}\tilde{D}(x) = 0$. The integrating factor $\rho(x)$ satisfies $\rho' = \rho(1 + \frac{1}{x-1})$ so $\rho(x) = (x-1)e^x$ will do. Hence $\frac{d}{dx}((x-1)e^x y) = 0$. Thus $(x-1)e^x y = C$ for some constant C . Comparing constant terms of both sides, we see $C = -1$ and $y = \frac{e^{-x}}{1-x}$.

$$e^{-x} \frac{1}{1-x} = \left(\sum_{j=0}^{\infty} \frac{(-1)^j}{j!} x^j \right) \left(\sum_{l=0}^{\infty} x^l \right)$$

so $d_n = n! \sum_{j=0}^n \frac{(-1)^j}{j!} = \sum_{j=0}^n (-1)^j \binom{n}{j} (n-j)!$. To interpret this using the principle of inclusion-exclusion, let $S(i)$ be the set of permutations of $[n]$ which fix the integer $i \in [n]$. Now d_n is the cardinality of $S_n - \cup_{i=1}^n S(i)$. Using inclusion-exclusion, it remains only to note that there are $\binom{n}{j}$ ways to intersect j distinct $S(i)$'s, and the cardinality of each of these j -fold intersections is $(n-j)!$.

6. For an integer j , one proves by induction that $a_{2j} = (2j-1)(2j-3)\dots 1$ as follows. We may take $[2j]$ to be our $2j$ -element set

and note that there are $2j - 1$ possibilities for what $2j$ pairs up with, say $i \in [2j - 1]$. By the inductive hypothesis, there are $(2j - 3)(2j - 5) \dots 1$ of splitting up the other $2j - 2$ so our formula for a_{2j} is proved. Now note

$$a_{2j} = (2j - 1)(2j - 3) \dots 1 = \frac{(2j)!}{2j(2j - 2) \dots 2} = \frac{(2j)!}{2^j j!}$$

so

$$\tilde{A}(x) = \sum_{j=0}^{\infty} \frac{1}{2^j j!} x^{2j} = e^{\frac{1}{2}x^2}$$

In the party, there are $\binom{n}{k}$ possibilities for which k people stay single, in which case, there are a_{n-k} ways for the others to pair up. This gives the relation (2). Now the exponential generating function for $\{b_n\}_{n \in \mathbb{N}}$ is $\tilde{B}(x) = e^x$ so $\tilde{C}(x) = \tilde{A}(x)\tilde{B}(x) = e^{x + \frac{1}{2}x^2}$.

7. If σ permutes $[n]$ cyclically, then there are $n - 1$ choices for where $\sigma(1)$ goes, then $n - 2$ choices for $\sigma(\sigma(1))$ and so forth giving $a_n = (n - 1)!$ in total if we interpret $(-1)! = 0$. Thus $\tilde{A}(x) = \sum_{n=1}^{\infty} \frac{1}{n} x^n$. Differentiating gives $\tilde{A}'(x) = \frac{1}{1-x}$ so integrating term by term and noting $a_0 = 0$ gives $\tilde{A}(x) = -\log(1 - x)$.

To find $\tilde{B}(x)$, suppose σ is a 2 orbit permutation of $[n]$ which cyclically permutes a k -element subset S_1 , and its $(n - k)$ -element complement. There are $\binom{n}{k}$ choices of S_1 , and a_k ways to cyclically permute S_1 , and a_{n-k} ways to cyclically permute S_2 . Naturally, we need to sum over $k = 1, \dots, n - 1$ (actually since $a_0 = 0$, we can even sum from 0 to n), but note this double counts what we want. For example, the σ above gets counted again when k is changed to $n - k$, and we swap the roles of S_1 and S_2 . Thus

$$c_n = \frac{1}{2} \sum_{j=0}^n \binom{n}{j} a_j a_{n-j}.$$

This is half what we find in (2) so $\tilde{B}(x) = -\frac{1}{2} \log(1 - x)$.

For j orbit permutations σ , we can apply the same procedure to find $S_1 \subset [n]$ on which σ permutes the elements cyclically, and so gives a $j - 1$ orbit permutation on the complement. Since there are j possibilities for S_1 , the overcount this time is by a factor of j and the above argument now gives $\tilde{C}_j(x) = \frac{1}{j} \tilde{A}(x) \tilde{C}_{j-1}(x) = \frac{1}{j!} (-\log(1 - x))^j$. Finally, note that

$$\begin{aligned} 1 + \tilde{A}(x) + \tilde{B}(x) + \tilde{C}_2(x) + \dots &= \\ 1 + (-\log(1 - x)) + \frac{1}{2!} (-\log(1 - x))^2 + \frac{1}{3!} (-\log(1 - x))^3 + \dots &= \\ &= e^{-\log(1-x)} = (1 - x)^{-1} \end{aligned}$$

Extension III Mathematics Activity Sheet Partitions

You might want to check out the teaser video on “Partitions” before trying this Activity Sheet.

A *partition* of $n \in \mathbb{N}$ is a sequence of positive integers $n_1, n_2, \dots, n_k$ which sums to n and is *weakly decreasing* in the sense that $n_1 \geq n_2 \geq \dots \geq n_k$. Here we say the partition has k parts, the parts being the n_i . It is interesting to count partitions satisfying various conditions, and generating functions often arise naturally.

For example suppose we consider only partitions whose parts are $\leq k$. Let a_n be the number of such partitions of n . The key is to consider partitions of all possible n at the same time, which may be enumerated as a_1 1's, a_2 2's, $\dots$, a_k k 's, where the a_i 's are arbitrary non-negative integers. More concisely, we consider the set of all k -tuples $\vec{a} = (a_1, \dots, a_k) \in \mathbb{N}^k$ which is in bijection with the set of all partitions (of all possible n) with (all) parts $\leq k$. We associate to this the monomial $m_{\vec{a}}(x_1, \dots, x_k) := x_1^{a_1} x_2^{a_2} \dots x_k^{a_k}$. The sum of all these can be collected in the product

$$\begin{aligned} P(x_1, x_2, \dots, x_k) &= \sum_{\vec{a} \in \mathbb{N}^k} m_{\vec{a}}(x_1, \dots, x_k) \\ &= (1 + x_1 + x_1^2 + \dots)(1 + x_2 + x_2^2 + \dots) \dots (1 + x_k + \dots) \\ &= \frac{1}{1 - x_1} \frac{1}{1 - x_2} \dots \frac{1}{1 - x_k} \end{aligned} \tag{1}$$

Now $\vec{a}$ corresponds to a partition of n if and only if $x^n = m_{\vec{a}}(x, x^2, \dots, x^k)$ so we get the generating function

$$\sum_{n=0}^{\infty} a_n x^n = P(x, x^2, \dots, x^k) = \prod_{j=1}^k \frac{1}{1 - x^j} \tag{2}$$

In this Activity Sheet, we will look at similar counting problems by considering other classes of partitions, and even looking at the case where k goes to infinity to study all partitions.

To do this Activity Sheet, you will need quite some familiarity with generating functions. This includes the material in the Activity Sheets on Counting Monomials and Formal Power Series.

Activities

- Find all the partitions of $n = 1, 2, 3, 4$. Calculate from definitions the coefficient of x^n in the series $P(x, x^2, x^3, x^4, x^5)$ above and explain why it gives you the number of partitions of n .
- Let $\{a_n\}_{n \in \mathbb{N}}$ be the number of partitions of n of the type below. Find the associated generating functions.
 - All parts are odd from $1, \dots, 2k + 1$.
 - All parts are $\leq k$ and no part is repeated more than l times i.e. at most l 1's, l 2's etc.
 - All parts are $\leq k$ and 1 appears an odd number of times.
- A Laurent series $A(x) = \sum_{n=d}^{\infty} a_n x^n$ with $a_d \neq 0$ is said to have *degree* $\deg A(x) = d$. By default, $\deg 0 = \infty$. Given a sequence of Laurent series $A_i(x), i \in \mathbb{N}$, we write $\lim_{i \rightarrow \infty} \deg A_i(x) = \infty$ if for any $d \in \mathbb{Z}$, only finitely many $A_i(x)$ have degree $\leq d$.
 - Given Laurent series $A(x), B(x)$, show that $\deg A(x)B(x) = \deg A(x) + \deg B(x)$.
 - Show that the condition $\lim_{i \rightarrow \infty} \deg A_i(x) = \infty$ is equivalent to saying that for any $d \in \mathbb{Z}$, there exists some $N \in \mathbb{N}$ such that $\deg A_i(x) > d$ for all $i \geq N$.
 - Assume for the rest of this question that $\lim_{i \rightarrow \infty} \deg A_i(x) = \infty$. Let s_{nN} be the coefficient of x^n in $\sum_{i=0}^N A_i(x)$. Show that the sequence $s_{n0}, s_{n1}, \dots$ is eventually equal to some constant, say s_n .
 - Hence we will say that $\sum_{i=0}^{\infty} A_i(x)$ *converges formally* and that the infinite *sum* is $\sum_n s_n x^n$. Recall the power series $\frac{1}{1-x} = 1 + x + x^2 + \dots$. Show that $\sum_{n=0}^{\infty} \frac{x^n}{1-x}$ converges formally, and find the sum.
- Suppose the sums $\sum_i A_i(x)$ and $\sum_i B_i(x)$ both converge formally to $A(x)$ and $B(x)$ respectively. We define the Cauchy product series to be $\sum_{n=0}^{\infty} C_n(x)$ where as expected $C_n(x) = \sum_{i+j=n} A_i(x)B_j(x)$. Show that the Cauchy product series converges formally to $A(x)B(x)$.
- Redo Q3d) using the previous question.

6. Let $A(x) = a_0 + a_1x + \dots$ be a power series with $a_0 \neq 0$. We know $A(x)$ is invertible in $\mathbb{C}[[x]]$ and here give an interesting proof which also provides a formula for the inverse, though not always a useful one.
 - (a) Show that it suffices to prove the case when $a_0 = 1$.
 - (b) Suppose $a_0 = 1$ so $A(x) = 1 - B(x)$ where $B(x)$ is a power series with degree > 0 . Can you guess a formally convergent series which gives a formula for the inverse of $A(x)$? Hint: formally you want $\frac{1}{1-B(x)}$ which looks like a formula you see from the theory of geometric progressions.
 - (c) Prove your formula for the inverse.
7. Suppose now that $A_i(x), i \in \mathbb{N}$ is a sequence of Laurent series. Can you come up with a sensible condition on the sequence so that it makes sense to form the infinite product $\prod_{i \in \mathbb{N}} A_i(x)$?
8. For any positive integer i we define

$$A_i(x) = 1 + x^i - x^{i+1} + x^{2i} - x^{2i-1} + x^{3i} - x^{3i-1} + \dots$$

Compute $\prod_i A_i(x)$.

9. Let a_n be the number of partitions of n . Show that the corresponding generating function is $\prod_{i=1}^{\infty} \frac{1}{1-x^i}$.
10. Here we look at Euler's classic proof that the number a_n of partitions of n into distinct parts is equal to the number b_n of partitions of n into odd parts. Let $A(x), B(x)$ be their corresponding generating functions.
 - (a) Find all the partitions of 5 into odd parts, and into distinct parts.
 - (b) Find expressions for $A(x), B(x)$ as infinite products.
 - (c) Prove that $A(x) = B(x)$ and hence that $a_n = b_n$ for all n .

Partial answers, comments and hints

1. $2 = 1 + 1, 3 = 2 + 1 = 1 + 1 + 1, 4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$. Any partition of $n < 5$ has largest part < 5 so the coefficient of x^n in $P(x, x^2, x^3, x^4, x^5)$ certainly will give the number of partitions of 3 and can be calculated as below using the fact that products with terms of degree > 4 will stay of degree > 4 .

$$\begin{aligned}
& (1+x+x^2+x^3+x^4+\dots)(1+x^2+x^4+\dots)(1+x^3\dots)(1+x^4+\dots)(1+\dots) \\
&= (1+x+2x^2+2x^3+3x^4+\dots)(1+x^3\dots)(1+x^4+\dots) \\
&= (1+x+2x^2+3x^3+4x^4+\dots)(1+x^4+\dots) \\
&= 1+x+2x^2+3x^3+5x^4+\dots \quad (3)
\end{aligned}$$

The number of partitions of 0, 1, 2, 3, 4 is 1, 1, 2, 3, 5, the first few coefficients of $P(x, x^2, x^3, x^4, x^5)$.

2. For a) $\prod_{j=0}^k \frac{1}{1-x^{2j+1}}$ retaining only GPs with odd powers of x as common ratios. For b), the infinite GPs $1+x_i+x_i^2+\dots$ in (1) must be replaced by finite ones since all $a_i \leq l$ giving $\prod_{j=1}^k \frac{1-x^{j(l+1)}}{1-x^j}$. Finally, for c), the GP $1+x_1+x_1^2+\dots$ in (1) above must be replaced by $x_1+x_1^3+x_1^5+\dots$ giving $\frac{x}{1-x^2} \prod_{j=2}^k \frac{1}{1-x^j}$.
3. Parts a)- c) are hopefully straightforward. For example, assuming the result and notation in b), we see the coefficient of x^d in $A_i(x)$ is zero for all $i \geq N$ so $s_{dN} = s_{d,N+1} = s_{d,N+2} = \dots$ and c) follows.

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{x^n}{1-x} &= 1+x+x^2+\dots \\
&+ x+x^2+\dots \\
&+ x^2+\dots \\
&\vdots \\
&= 1+2x+3x^2+\dots = \frac{1}{(1-x)^2}
\end{aligned}$$

Of course, this is the same answer as you would get if you pulled out $\frac{1}{1-x}$ from each term and assumed an “infinite” distributive law worked. You should check that this is indeed valid or see the following questions.

4. Both parts use similar arguments. We will assume the sum $\sum C_i(x)$ converges, say to $C(x)$ and just prove $C(x) = A(x)B(x)$. We need only show that for any fixed n , the coefficients of x^n on both sides are the same. Firstly, we may assume that there is a sufficiently negative integer M such that the coefficients of x^j in any of the A_i 's or B_i 's is zero whenever $j < M$. We may also assume that N is a sufficiently positive integer that $\deg A_j(x), \deg B_j(x) > n - M$ whenever $j \geq N$. Thus $\Delta_A(x) := A(x) - \sum_{i=0}^N A_i(x)$ and $\Delta_B(x) := B(x) - \sum_{i=0}^N B_i(x)$ both have degrees $> n - M$. It follows that

$$\begin{aligned} A(x)B(x) - \sum_{i=0}^N A_i(x) \sum_{i=0}^N B_i(x) = \\ \left(\sum_{i=0}^N A_i(x) \right) \Delta_B(x) + \Delta_A(x) \left(\sum_{i=0}^N B_i(x) \right) + \Delta_A(x) \Delta_B(x) \end{aligned}$$

also has degree $> n$. The coefficient of x^n in $A(x)B(x)$ is the same as that in $\sum_{i=0}^N A_i(x) \sum_{i=0}^N B_i(x)$. Repeating this argument, we see that this also has the same coefficient of x^n as

$$\sum_{i+j \leq N} A_i(x) B_j(x) = \sum_{i=0}^N C_i(x).$$

Arguing similarly, we see that this also has the same coefficient of x^n as $C(x)$ and we are done.

5. Take the Cauchy product of $1 + x + x^2 + \dots = \frac{1}{1-x}$ and $\frac{1}{1-x} + 0 + 0 + \dots$
6. Suppose $C(x)$ is an inverse to $a_0^{-1}A(x) = 1 + \dots$. Then $a_0^{-1}C(x)$ is the inverse of $A(x)$. Thus we may assume $a_0 = 1$. Consider the sum $1 + B(x) + B(x)^2 + \dots$ which clearly converges formally. The Cauchy product with $1 - B(x)$ is 1 so from Q4 we see

$$(1 - B(x))^{-1} = 1 + B(x) + B(x)^2 + \dots$$

7. For any positive integer d , let $1 + x^d \mathbb{C}[[x]]$ be the set of power series of the form $1 + B(x)$ where $\deg B(x) \geq d$ or equivalently, we can

factorise $B(x) = x^d C(x)$ for some $C(x) \in \mathbb{C}[[x]]$. The condition for formal convergence of $\prod_i A_i(x)$ is that for any such d , all but finitely many of the $A_i(x)$ lie in $1 + x^d \mathbb{C}[[x]]$. This condition ensures that all but finitely many of the Laurent series are actually power series, so in our analysis, we may as well assume all the $A_i(x) \in \mathbb{C}[[x]]$. The key point is that given $B(x) \in 1 + x^d \mathbb{C}[[x]]$ and any $D(x) \in \mathbb{C}[[x]]$, the coefficients of $1, x, \dots, x^{d-1}$ are the same for both $D(x)$ and $D(x)B(x)$. Thus if we pick N large enough so that $A_i(x) \in 1 + x^d \mathbb{C}[[x]]$ for all $i \geq N$, then all the partial products $\prod_{i=1}^n A_i(x)$ will have the same coefficients of $1, x, \dots, x^{d-1}$ as soon as $n \geq N$. We can thus use them to define the coefficients of $1, x, \dots, x^{d-1}$ in the infinite product $\prod_{i=1}^{\infty} A_i(x)$.

8. Sum the two geometric series to see $A_i(x) = \frac{1-x^{i+1}}{1-x^i}$ so the partial products telescope in the sense that

$$A_1(x)A_2(x)\dots A_k(x) = \frac{1-x^2}{1-x} \frac{1-x^3}{1-x^2} \dots \frac{1-x^{k+1}}{1-x^k} = \frac{1-x^{k+1}}{1-x}$$

so the infinite product is just $\frac{1}{1-x} = 1 + x + x^2 + \dots$

9. First note that the infinite product converges formally. Also, any partition of n has all parts $\leq n$, so for any $N \geq n$, the coefficient of x^n in the partial product $\prod_{i=1}^N \frac{1}{1-x^i}$ will be a_n .
10. Partitions of 5 into odd parts are $5 = 3 + 1 + 1 = 1 + 1 + 1 + 1 + 1$ whilst partitions into distinct parts are $5 = 4 + 1 = 3 + 2$.

$$A(x) = \prod_{i=1}^{\infty} \frac{1}{1-x^{2i-1}}, \quad B(x) = \prod_{i=1}^{\infty} (1+x^i).$$

But

$$B(x) = \prod_{i=1}^{\infty} (1+x^i) = \prod_{i=1}^{\infty} \frac{1-x^{2i}}{1-x^i} = A(x)$$

on cancelling all the $1-x^{2i}$ terms in the numerator with the corresponding term in the denominator. If you think this last step is fishy, it's because technically, one isn't done yet. You need to justify why you can do it.

Extension III Mathematics Activity Sheet

Difference Operator

Do watch the associated video “Difference Operator” before doing this Activity Sheet. A quick summary of some of the main points is below.

We view sequences as functions $f: \mathbb{N} \rightarrow \mathbb{C}$ (or from $\mathbb{Z} \rightarrow \mathbb{C}$ in the doubly infinite case). One can add two such sequences f, g pointwise, so $(f + g)(n) := f(n) + g(n)$ or multiply by a scalar $c \in \mathbb{C}$ so $(cf)(n) := cf(n)$. The *difference operator* Δ takes any sequence f and outputs a new sequence Δf defined by

$$(\Delta f)(n) := f(n+1) - f(n)$$

Eg If $p_k(n) = \binom{n}{k}$ then $\Delta p_{k+1} = p_k$ for $k \in \mathbb{N}$.

One should think of Δ as a *discrete* version of the differentiation operator, and as such, many of the usual properties for differentiation have their discrete analogues. The aim of this Activity Sheet is for you to look at some of these properties and see how they can be used to study sequences in much the same way that the derivative can be used to study functions of a real-variable. We will also see how it is linked to the concept of generating functions, so ideally, you should be familiar with the material in the Activity Sheets on Recurrence Relations and Generating Functions, but that is about all you'll need to know to do this Activity Sheet.

Activities

1. Consider an arithmetic progression $f(n) = a + nd$ for some fixed constants $a, d \in \mathbb{C}$. What is the sequence Δf ?
2. Like the differentiation operator, Δ is *linear* in the sense that $\Delta(f + g) = \Delta f + \Delta g$ for sequences f, g and if $c \in \mathbb{C}$ we also have $\Delta(cf) = c(\Delta f)$. Prove this.
3. Show that $\Delta f = 0$ (the right hand side denotes the constant sequence with value 0), if and only if f is a constant sequence. Show that $\Delta f = \Delta g$ if and only if $f - g$ is constant.
4. Is there a discrete analogue of the Fundamental Theorem of Calculus, and if so, how does one prove it?

5. Show that any polynomial $p(n)$ of degree d can be written in the form

$$p(n) = a_0 p_0(n) + a_1 p_1(n) + \dots + a_d p_d(n)$$

for some constants $a_0, \dots, a_d$ and $p_i(n)$ as in the Example above.

6. Find a closed formula for the sequence $\Sigma^3(n) := 1^3 + 2^3 + \dots + (n-1)^3$. (A closed formula is one without “...” in the formula and instead involves a fixed number of operations not depending on n).
7. Show that for any $k \in \mathbb{Z}_+$, the sequence $\Sigma^k(n) := 1^k + 2^k + \dots + (n-1)^k$ is a polynomial function of n of degree $k+1$.
8. Just as we can take second and higher derivatives, one can apply the difference operator successively k times to get the operator Δ^k . Show that the sequences f such that $\Delta^2 f = 0$ are precisely the arithmetic progressions, i.e. linear functions.
9. Show that if $p(n)$ is a polynomial function of n of degree d , then $\Delta^{d+1} p = 0$.
10. Let $F(x) = \sum_{n=0}^{\infty} f(n)x^n$ be the generating function for a sequence f . What is the generating function of Δf in terms of $F(x)$.
11. Show that $\Delta^{d+1} f = 0$ if and only if the generating function $F(x)$ of the sequence f has the form $F(x) = \frac{p(x)}{(1-x)^{d+1}}$ for some polynomial p of degree $\leq d$.
12. Show that the following conditions on a sequence f are equivalent.
- (a) f is defined by a polynomial of degree $\leq d$.
 - (b) The corresponding generating function $F(x)$ has the form $\frac{p(x)}{(1-x)^{d+1}}$ for some polynomial of degree $\leq d$.
 - (c) $\Delta^{d+1} f = 0$.

Partial answers, comments and hints

1. $\Delta f = d$, the constant function with value d .

2. Just follow the definitions and show functions on both sides have the same values for all $n \in \mathbb{N}$. For example

$$\begin{aligned} [\Delta(cf)](n) &= (cf)(n+1) - (cf)(n) = cf(n+1) - cf(n) \\ &= c[f(n+1) - f(n)] = c(\Delta f)(n) \end{aligned}$$

3. This should be clear, note that the second part follows from the first and linearity proved in Q2.
4. From, Q3, one can at best invert the difference operator up to the analogue of the integration constant. The desired “inverse” operation is of course, taking partial sums. More precisely, if we have the relation $\Delta f = g$, then one discrete form of the Fundamental Theorem of Calculus is

$$g(a) + g(a+1) + \dots + g(b-1) = f(b) - f(a)$$

which is easily seen by observing the LHS is a telescoping sum. The discrete analogue of the other part of the Fundamental Theorem of Calculus is, given a sequence g and constant $C \in \mathbb{C}$, if we define a new sequence f via partial sums

$$f(n) = C + g(0) + g(1) + \dots + g(n-1),$$

then $\Delta f = g$ and in fact, f is the unique function which has this property and satisfies the initial condition $f(0) = C$.

5. This is by induction on degree, the case $d = 0$ being clear. Suppose $p(n)$ has degree d say with leading term $b_d n^d$. Now $p_d(n) = \frac{1}{d!} n(n-1)\dots(n-d+1)$ has leading term $\frac{1}{d!} n^d$. Let $a_d = d!b_d$ so that $p(n) - a_d p_d(n)$ is a polynomial of degree $< d$ and so, by induction

$$p(n) - a_d p_d(n) = a_0 p_0(n) + a_1 p_1(n) + \dots + a_{d-1} p_{d-1}(n)$$

for some constants $a_0, \dots, a_{d-1}$ as desired.

6. $(\Delta \Sigma^3)(n) = n^3$ which we wish to write in the form $a_0 p_0 + a_1 p_1 + a_2 p_2 + a_3 p_3$. We can follow the algorithm implicit in the proof of Q5 to see $n^3 = 6\binom{n}{3} + 6\binom{n}{2}\binom{n}{1}$ so using the Example above and linearity we see

$$\Sigma^3(n) = 6\binom{n}{4} + 6\binom{n}{3}\binom{n}{2} = \frac{1}{4}n^2(n-1)^2.$$

7. This is just a generalisation of the previous question.
8. $\Delta(\Delta f) = 0$ if and only if Δf is the constant function say d by Q3. But $\Delta f = d$ just means that f defines an arithmetic progression with constant difference d .
9. You can prove this directly or use Q5 to rewrite $p(n) = a_0p_0(n) + a_1p_1(n) + \dots + a_dp_d(n)$. Then the example in the intro and linearity (see Q2) show

$$\Delta^{d+1}p = \Delta^d(\Delta p) = \Delta^d(a_1p_0 + \dots + a_dp_{d-1}) = \dots = 0.$$

10. The generating function is $(x^{-1} - 1)F(x) - F(0)x^{-1}$.
11. We can interpret the answer to Q10 as, the generating function of Δf is gotten from the generating function $F(x)$ of f by multiplying by $x^{-1} - 1$ and dropping the x^{-1} term. Thus the generating function of $\Delta^{d+1}f$ is gotten from $(x^{-1} - 1)^{d+1}F(x)$ by dropping all the terms involving $x^{-d-1}, x^{-d}, \dots, x^{-1}$. Thus $\Delta^{d+1}f = 0$ means $(x^{-1} - 1)^{d+1}F(x) = p_0x^{-d-1} + p_1x^{-d} + \dots + p_dx^{-1}$. Multiplying through by x^{d+1} shows $(1 - x)^{d+1}F(x) = p(x)$ for some polynomial p of degree $\leq d$.
12. The equivalence of (b) and (c) has already been established in Q11. In Q9, we also saw (a) implies (c). It remains only to show that (b) or (c) implies (a). This can be done in many ways. One way is to use the partial fraction expansion of $F(x)$ which reduces the question to showing that the coefficients of $\frac{1}{(1-x)^{k+1}}$ is a polynomial sequence of degree k . Now

$$\begin{aligned} \frac{1}{(1-x)^{k+1}} &= (-1)^k \frac{1}{k!} \frac{d^k}{dx^k} \left(\frac{1}{1-x} \right) = (-1)^k \frac{d^k}{dx^k} \left(\sum_{n=0}^{\infty} \frac{1}{k!} x^n \right) \\ &= (-1)^k \sum_{n=0}^{\infty} \binom{n}{k} x^{n-k} = (-1)^k \sum_{n=0}^{\infty} \binom{n+k}{k} x^n \end{aligned}$$

and $\binom{n+k}{k}$ is a polynomial function of n of degree k .